

Reviewer(s)' Comments to Author:

Reviewer: 1

Comments to the Author

1. All abbreviations should be defined at their first appearance in the manuscript.
2. The authors should provide a more detailed description of the geographic unit used for exposure assignment and linkage to individual participants.
3. For clarity, the specific gestational periods corresponding to each trimester should be explicitly described.
4. The authors should justify the use of four binary exposure variables. Dichotomizing exposure may result in a substantial loss of information and potentially reduce the ability to detect exposure–response relationships.
5. In addition, the definitions of the "threat," "impact," and "cleanup" phases are somewhat ambiguous. The authors should provide clearer operational definitions and explain how these exposure windows were determined.
6. The process used to select covariates should be clearly justified, preferably based on prior literature, a conceptual framework, or a directed acyclic graph (DAG).
7. Maternal and paternal education as well as race/ethnicity may be highly correlated. Did the authors assess potential multicollinearity among these variables? If so, the results should be reported.
8. Beyond sociodemographic characteristics, could environmental factors such as air pollution and meteorological conditions be considered as potential confounders?
9. Exposure defined as any tropical cyclone (TC) experience during pregnancy may represent a relatively crude exposure metric. The potential for exposure misclassification should be discussed in greater detail.
10. If TC exposure is concentrated within specific seasons or months, it may be more appropriate to restrict analyses to pregnancies that overlapped with periods when TC exposure was realistically possible. The current exposure definition may introduce substantial heterogeneity in exposure opportunity across pregnancies.
11. Furthermore, the authors do not appear to account for seasonality or long-term temporal trends. In my view, adjustment for temporal patterns is a critical consideration in this study. Without adequate control for season and calendar year, the observed associations may partially reflect underlying temporal trends, such as concurrent increases in TC occurrence and changes in the incidence of low birth weight over the study period, rather than a true effect of TC exposure itself.

Reviewer: 2

Comments to the Author

Overall Assessment

This manuscript presents a comprehensive, large-scale retrospective cohort study (N = 4,514,346) investigating the impact of tropical cyclone exposure during pregnancy on a broad range of adverse maternal and neonatal health outcomes in Florida over a 22-year

period (2000–2022). Although the topic is important and the dataset is impressive, the study is fundamentally limited by simplistic exposure assessment, limited analytical depth, and concerns regarding causal interpretation. Many of these issues are intrinsic to the study design and are unlikely to be fully resolved through revision. Therefore, I do not recommend publication in JESEE.

Major concerns

1. Exposure definition requires stronger justification and sensitivity analyses. The primary exposure classification may not adequately reflect the complex and prolonged nature of TC related disruptions. Defining the recovery phase as only one day after landfall appears arbitrary and lacks a clear epidemiologic or biological rationale. The authors should provide stronger justification for these exposure windows and, ideally, conduct sensitivity analyses using alternative exposure durations (e.g., ± 3 days, ± 7 days, or longer post-landfall periods) to evaluate the robustness of the findings.
2. Lack of exposure intensity assessment. All exposure variables are modeled as binary indicators. The manuscript notes that information on storm category and wind characteristics was available, yet these variables were not incorporated into the analysis. The authors should consider additional analyses evaluating exposure intensity, such as storm category, maximum wind speed, storm size, or cumulative exposure metrics.
3. Potential temporal confounding over the 22-year study period. The statistical models do not appear to adjust for calendar year or temporal trends. Because both TC activity and pregnancy outcomes may vary over time, failure to account for secular trends could introduce residual confounding.
4. Potential spatial confounding. Tropical cyclone exposure is inherently spatially clustered. Women residing in coastal areas are substantially more likely to experience cyclone exposure than those living inland. Coastal and inland populations may differ in socioeconomic status, healthcare access, environmental conditions, and baseline health risks. The authors should discuss this limitation more thoroughly and consider incorporating geographic controls.
5. The reported protective associations during the second trimester warrant careful interpretation. These findings may reflect live-birth bias or competing-risk selection bias. The authors should explicitly acknowledge and discuss the potential influence of live-birth bias and competing-risk selection bias in the Discussion section.
6. Interpretation of extremely small effect sizes. The observed statistically significant associations may have limited public health relevance. The authors should provide a more cautious interpretation of these small effect sizes and consider the potential influence of residual confounding.
7. Mechanistic discussion is limited. The discussion focuses primarily on prenatal maternal stress as a potential mechanism. While stress is certainly plausible, tropical cyclones may affect pregnancy outcomes through numerous pathways, including disruptions in prenatal care, medication access, nutrition, environmental exposures, infectious disease risk, displacement, and prolonged power outages. A broader discussion of potential biological and social mechanisms would strengthen the manuscript.
8. The handling of missing data warrants further consideration. The manuscript does not

discuss whether the missing-at-random assumption underlying multiple imputation is plausible. In addition, sensitivity analyses are recommended to evaluate the robustness of the findings to the multiple imputation procedure.

Minor concerns

1. The title may not fully reflect the scope of the study. The term "adverse pregnancy outcomes" may not adequately convey the inclusion of maternal complications such as GDM and HDP.

Reviewer: 3

Comments to the Author

General comments

This is a well-intentioned study using a large, valuable dataset to examine an important public health question. However, I have several methodological concerns that limit my confidence in the findings. The exposure classification approach, i.e., treating a 3-day window (day before, day of, day after landfall) as "threat, impact, and cleanup", strikes me as overly simplistic and unlikely to capture the true nature of TC-related stress. Additionally, the numerous protective associations (e.g., decreased odds of vPTB, vLBW) are not adequately explained and raise questions about residual confounding or exposure misclassification. The authors should consider whether these findings represent real biological effects or methodological artifacts.

Specific comments

Methods

Lines 121-145: I am concerned about the exposure definition. The authors classify TC exposure based solely on whether a birth location falls within the storm's wind swath on three specific days (before, during, after landfall). This seems to miss the broader experience of a TC, prolonged power outages, flooding, disrupted healthcare access, evacuation stress, property damage, that may extend for weeks. Please explain why you believe a 3-day window adequately captures TC-related stress during pregnancy.

Lines 140-145: The distinction between "threat," "impact," and "cleanup" appears arbitrary to me. In reality, these phases often overlap and are experienced differently depending on individual circumstances. Moreover, using birth residence to assign exposure assumes the mother lived at that location throughout the relevant trimester. Given evacuation and displacement are common during TCs, I worry this creates substantial exposure misclassification.

Lines 170-181: The authors adjust for race/ethnicity and education but not for area-level factors (e.g., county-level poverty, urbanicity, storm preparedness infrastructure) that could confound the association. I would recommend the authors consider multi-level modeling or include contextual covariates.

Lines 173-177: Multiple imputation with 20-34% missingness for key covariates (race, BMI) is appropriate, but the authors do not describe whether missingness was differential by exposure status. This matters because storm-affected areas might have different data completeness patterns. I suggest the authors provide a missingness analysis by exposure group.

Lines 192-196: The decreased odds of vPTB and vLBW in the second trimester (aOR=0.919) are puzzling. If TC exposure is stressful, why would it reduce extreme preterm birth? The authors speculate about altered behaviors or differential biological pathways, but this does feel speculative. Please provide a more compelling explanation or acknowledge this as a potential artifact of exposure misclassification or selection bias.

Lines 196-200: Similar protective associations for tLBW in the third trimester (aOR 0.966) require careful interpretation. I wonder if this reflects selective survival, pregnancies already at risk may have delivered earlier, leaving a healthier cohort exposed later. Please address this possibility directly.

Lines 201-203: The GDM (aOR 1.05) and HDP (aOR 1.127) associations are interesting but the effect sizes are modest. With 4.5 million births, even tiny associations reach statistical significance. I question whether a 5% increase in GDM odds is clinically meaningful, particularly given the exposure is defined at the trimester level.

Lines 215-218: I appreciate the cautious interpretation of decreased odds, but the explanation remains vague. The phrase "altered behaviors in anticipation of TC events" needs more specificity.

Lines 237-254: The authors correctly note that exposure definitions vary across studies. However, I feel they overstate the novelty of their approach. Using spatially joined track data is an improvement over county-level or disaster declaration methods, but the 3-day exposure window still seems limited.

Lines 255-260: The authors honestly acknowledge that birth residence may misclassify exposure due to displacement. This is a major limitation that I believe deserves more prominent discussion. If mothers evacuated, they might have been exposed to a different environment than the one assigned. Authors should consider sensitivity analyses restricting to areas where evacuation was less likely or among populations less likely to relocate.

Reviewer: 4

Comments to the Author

This manuscript examines the association between tropical cyclones (TCs) and adverse pregnancy outcomes in a large retrospective cohort study in Florida using birth certificate data from 2000–2022. The analysis is rigorous and the manuscript provides a detailed description of how the timing of TC exposure during pregnancy may influence risk of adverse birth outcomes. This study contributes to a literature base that has reported mixed

findings to date.

Major:

1. Although multiple imputation is an appropriate method for handling missing data, the proportion of missingness for some variables is substantial (over 20-30% for some key covariates). I encourage the authors to provide additional information regarding the imputation procedure (e.g., variables included in the imputation model, how number of imputations was selected, diagnostics) and to consider presenting a complete-case sensitivity analysis to demonstrate whether the conclusions are robust to alternative approaches for handling missing data.

2. The manuscript would benefit from a clearer articulation of the hypothesized causal pathways linking tropical cyclone exposure to adverse pregnancy outcomes. In addition to maternal psychological stress, potential mechanisms may include physical injury, increased risk of communicable disease transmission, disruptions in healthcare access, and indirect effects such as power outages affecting refrigeration, medication use, and environmental conditions. Providing a more detailed discussion of these pathways would strengthen the interpretation of the findings. The inclusion of a conceptual framework or directed acyclic graph (DAG) would further improve clarity regarding assumed relationships and potential sources of confounding and mediation.

Minor

1. The manuscript would benefit from additional detail regarding the residential geolocation data. Specifically, please clarify how latitude and longitude were determined and whether these coordinates correspond to the maternal residence reported at delivery, during pregnancy, or at another time point.

2. Please report the proportion of birth records for which accurate geocoded latitude and longitude information was unavailable, as well as any exclusions resulting from missing or low-quality geocoding data.

3. Please provide additional information regarding the definition and ascertainment of race and ethnicity variables. For example, how were categories defined for analysis?

4. Additional clarification of the exposure definition (e.g., spatial buffers, timing windows, or storm intensity metrics) would improve interpretability. Can the authors please expand on the selection of storms based on wind speeds, and the decision to buffer the tracts using the RMW?

5. The P values presented in Table 1 do not appear to add meaningful information and could be removed. Current epidemiologic reporting guidelines generally discourage significance testing of baseline characteristics, as these comparisons do not inform confounder selection and may overemphasize trivial differences in large samples. Consider presenting only the descriptive characteristics of the study population.

Hayes-Larson E, Kezios KL, Mooney SJ, Lovasi G. Who is in this study, anyway? Guidelines for a useful Table 1. *J Clin Epidemiol*. 2019 Oct;114:125-132. doi: 10.1016/j.jclinepi.2019.06.011. Epub 2019 Jun 20. PMID: 31229583; PMCID: PMC6773463.

6. Given well-described heterogeneity in vulnerability to extreme weather events, the authors may also consider whether effect modification by socioeconomic status or race/ethnicity is present.

Reviewer: 5

Comments to the Author

This is a large-scale retrospective cohort study that makes a meaningful contribution to the growing literature on climate-related hazards and perinatal health. The authors leverage an impressive sample of over 4.5 million Florida births across 23 years and employ a more nuanced exposure classification than many previous studies by distinguishing threat, impact, and cleanup phases of tropical cyclones. The study has substantial strengths, but several methodological and interpretive issues need to be addressed to strengthen the validity and clarity of the findings.

Specific Comments:

1. The authors define exposure based on residential address at birth being within the TC's effect zone during each trimester. This raises a critical concern: the analytic sample is conditional on delivering in Florida, and individuals who were exposed to a TC and subsequently relocated out of state (either temporarily or permanently) would be excluded from the birth records. If exposure is associated with out-migration, particularly among those with greater socioeconomic resources, this could introduce collider stratification bias and produce associations that do not reflect the true causal effect of TC exposure.
2. The authors observe decreased odds of vPTB, vLBW, and tLBW in certain trimesters and describe these as "should be interpreted cautiously." However, the discussion of potential mechanisms is relatively brief. These protective associations are striking and deserve more thorough consideration.
3. The authors note that threat, impact, and cleanup often co-occur (particularly when TCs occur near trimester boundaries), yet they present results for each exposure type separately across trimesters. This raises concerns about collinearity and redundant analyses. If these exposure categories are nearly perfectly correlated, presenting them side-by-side may overstate the distinctiveness of each phase.
4. The missingness rates for key covariates are substantial (e.g., mother's race 22.2%, father's race 33.6%, pre-pregnancy BMI 20%). While multiple imputation by chained equations is appropriate, the authors do not report:
 - a. How many imputed datasets were combined
 - b. Whether they used predictive mean matching or other methods for continuous variables.
 - c. Whether the missingness was associated with exposure or outcome (which could affect the validity of the imputation).

Please provide these details in the Methods section and consider including a table

comparing the distribution of covariates before and after imputation.

5. The exposure definition includes the day before, day of, and day after landfall. However, TC impacts (rainfall, wind, storm surge) can extend well beyond these time windows, and prenatal stress may persist for days or weeks after the event. The cleanup period as defined (day after landfall) may not capture the prolonged recovery period that could affect pregnant individuals. The authors should justify why they chose this narrow cleanup window and discuss how longer-term exposure or cumulative TC exposure across the pregnancy might be examined in future work. This is particularly relevant given the finding that "exact timing didn't make a difference", if the window is too narrow to capture meaningful variation, this finding may be less informative.

6. Some confidence intervals for PTB in the second trimester barely exclude the null (aOR 1.008, 95% CI 1.000, 1.016). The authors describe these as "increased odds," which is technically correct, but the clinical significance of such a small effect should be discussed more critically. With a sample size of 4.5 million, small effects can achieve statistical significance. The authors should consider discussing minimum detectable effects or clinical versus statistical significance.

7. The authors note that Florida's resilience may differ from other Gulf Coast states, and they appropriately limit their generalizability claims. However, they might also consider that Florida has a unique hurricane season (June to November) and that seasonal effects on pregnancy outcomes (e.g., due to heat, nutrition, or infectious diseases) could confound or interact with TC exposure. Consider whether seasonality could be a confounder or effect modifier, and discuss whether the temporal distribution of TCs across the year might influence the results.

Reviewer: 6

Comments to the Author

This study of tropical storms and birth outcomes provides a solid contribution to an area with limited research. It is generally well done, though there are some potential issues that should be addressed.

Given the issues of gestational diabetes and hypertension classification from birth records (many), these analyses may be better suited to be supplemental or dropped. This would also allow for more focus on the preterm and low birth weight outcomes.

I would suggest the authors not use statistical significance testing, but rather focus on the patterns of effect estimates. I think you largely do a good job of this already, simply that there's no need to highlight statistical significance. Also the differences between the "significant" and "non-significant" effects are largely negligible.

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: context, process, and purpose. *The American Statistician*, 70(2), 129-133.

It is not clear in the methods anything was done to address the potential for fixed cohort bias (see Neophytou 2021) in the analyses of preterm birth outcomes.

Neophytou, A. M., Kioumourtzoglou, M. A., Goin, D. E., Darwin, K. C., & Casey, J. A. (2021). Educational note: addressing special cases of bias that frequently occur in perinatal epidemiology. *International journal of epidemiology*, 50(1), 337-345.

Table 1 should include the numbers of outcomes, and ideally should have demographic breakdowns by outcome status, along with exposure status. This is particularly important given that the numerators and denominators will change with every outcome examined here. Relatedly, for the very PTB and low birth weight outcomes, was the referent population the same as the full outcome classification or any birth without those outcomes (i.e., at-risk population). For example, were PTBs above 32 weeks included in denominator of the vPTB analysis?

p-values should not be included in table 1, the analysis is not set up to test differences in this way and these are likely confounded. See Westreich & Greenland 2013.

Westreich, D., & Greenland, S. (2013). The table 2 fallacy: presenting and interpreting confounder and modifier coefficients. *American journal of epidemiology*, 177(4), 292-298.

Given the long study period, was an evaluation of temporal trends?

As many of the covariates are noted as “precision variables”, it would be good to present results with a parsimonious model (only necessary confounders).

Table 3 needs to be formatted to fit on a page.

Reviewer: 7

Comments to the Author

This manuscript examines the association between tropical cyclone exposure during pregnancy and adverse birth outcomes. The topic is timely and important, especially given the increasing frequency and intensity of extreme weather events. However, several methodological details require clarification before the findings can be fully interpreted. Abstract : The phrase “effect zone” should also be revised to “affected zone” in the abstract and throughout the manuscript.

Methods:

1. Sample selection and study flow

The authors should provide a sample flow chart showing the number of records at each stage of inclusion and exclusion

2. Line 101: The authors should explain what the latitude and longitude data represent. Were these coordinates based on maternal residential address?

3. If the geocoding is at the residential level, whether ethical approval was obtained for this

study should be clearly stated.

4. Exposure assessment: The authors should clearly explain how tropical cyclone exposure was assigned to each birth record. The authors should also describe how trimester-specific exposure was defined. For example, were exposures assigned separately for the first, second, and third trimesters based on the estimated gestational timeline? How were births with incomplete gestational information handled?

5. Residential mobility during pregnancy: Lines 138–139 raise an important issue. The authors should clarify how they handled mothers who gave birth in Florida but may have lived in another state during the first or second trimester. If exposure was assigned only based on residence at delivery, potential exposure misclassification due to residential mobility should be discussed. If residential history was available, the authors should explain how it was incorporated.

6. Tropical cyclone definition

Lines 125–135 should provide more detail on the tropical cyclone exposure metrics. Tropical cyclones can also affect birth outcomes through heavy rainfall, flooding, storm surge, displacement, infrastructure disruption, and reduced access to healthcare. The authors should clarify whether rainfall and flooding indicators were considered. If not, this limitation should be acknowledged.

In addition, the missingness percentage for radius of maximum wind should be reported, and discuss the reliability of these exposure data.

7. Definition of threat, impact, and cleanup periods

Lines 141–145 introduce the terms “threat,” “impact,” and “cleanup,” but these terms are confusing without definition. The authors should provide brief definitions when these terms are first introduced in the abstract.

8. Multiple imputation

Lines 179–181 should include more details about the multiple imputation procedure. The authors should report the number of imputed datasets, variables included in the imputation model, and the assumed missingness mechanism (MAR or MCAR). A sensitivity analysis by complete-case analysis may be warranted.

9. Potential confounding by temperature

Lines 148–157 : Temperature may be associated with both tropical cyclone occurrence and adverse birth outcomes. Therefore, adjustment for ambient temperature may be necessary. The authors should clarify whether temperature was included in the model. If not, they should justify this decision and discuss the possibility of residual confounding.

10. Long-term trends and seasonality

Since the study period spans 2000 to 2022, long-term temporal trends and seasonality

should be accounted for in the statistical models. Birth outcomes, healthcare access, reporting practices, and cyclone frequency may vary over time. The authors should explain how they controlled for calendar year, month or season of conception/birth, and other temporal patterns. If these were not controlled for, the potential impact on the findings should be discussed.

Results

11. Table 3 should be reformatted for clarity.

12. Description of tropical cyclone events

The Results section should provide more information on the tropical cyclone events included in the analysis. The authors should report how many tropical cyclone events were identified during the study period, their distribution over time, and the number of births exposed during each relevant pregnancy window.

13. Characteristics of affected and unaffected